Mikhail Razakov

C++ / LLVM Systems Engineer

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I build systems and compiler components in C++23/C17/Rust: LLVM passes, memory-dependence analysis, CFG/SSA, graph algorithms, x86-64 code generation, and reproducible test oracles.

LLVM / C++23
Optimization pass, graph algorithms, CMake, clang-tidy, ASan/UBSan

1st of 49 · 104/104
Conservative PS-form memory-dependence analysis with an exact oracle

x86-64 pipeline
SSA-like IR, liveness, linear scan, iced-x86, and differential validation

EXPERIENCE

July 1 — August 31, 2026

MCST — compiler engineering intern

- Implemented an LLVM LICM pass in C++ with conservative checks for side effects, memory access, loop invariance, preheaders, and speculative safety.
- Built a Python harness with baseline/optimized execution, opt -verify-each, and explicit CHANGE/NO_CHANGE expectations.

2026

PS-form Memory Dependence Analyzer

- Implemented conservative analysis of parametric memory accesses; ranked 1st of 49 with 104/104 tests.
- Built an exact oracle for small domains, metamorphic tests, and reproducible WA/TL counterexamples.

2026

x86-64 Codegen & Register Allocation Lab

- Designed SSA-like IR, liveness, linear scan, and parity checks between an IR interpreter and generated machine code.

PROJECTS

PS-form Memory Dependence Analyzer

Ranked 1st of 49 with 104/104 tests, with reproducible wrong-answer and time-limit counterexamples.

x86-64 Codegen & Register Allocation Lab

A measurable pipeline with spill/reload metrics, code size, and reproducible result comparison.

UniversalToolchain / Wist2

The current exact test manifest passes with zero failures; interpreter/CIL parity, clean-consumer scenarios, and reproducible compiler/runtime contracts are verified.

SKILLS

C++ / toolchain
C++23, C17, CMake, Linux, clang-tidy, ASan/UBSan, GitHub Actions

Compiler analysis
CFG, dominance, SSA, SCC, dataflow, memory dependence, optimization passes

Code generation
x86-64, iced-x86, liveness, linear scan, IR interpretation, isolated execution

Verification
exact oracles, differential/metamorphic tests, stress testing, counterexample reduction

EDUCATION

Software Engineering student at HSE University

RECOGNITION

The LangDev 2026 Program Committee accepted a talk on the architecture of UniversalToolchain/Wist2 for presentation in Torremolinos, Spain.

Moscow / remote